Prediction of Therapeutic Peptide using Deep Learning

PROJECT REPORT

Submitted by

KS Venkatram	**(CB.SC.U4AIE23236)**
Sanggit Saaran KCS	**(CB.SC.U4AIE23247)**
Vishal Seshadri B	**(CB.SC.U4AIE23260)**
Surya HA	**(CB.SC.U4AIE23267)**

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AMRITA VISHWA VIDYAPEETHAM

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COMPUTER SCIENCE ENGINEERING - ARTIFICIAL INTELLIGENCE

AMRITA VISHWA VIDYAPEETHAM

COIMBATORE - 641 112

BONAFIDE CERTIFICATE

This is to certify that the thesis entitled **"Prediction of Therapeutic Peptide using Deep Learning"** submitted by **KS Venkatram (CB.SC.U4AIE23236), Sanggit Saaran KCS (CB.SC.U4AIE23247), Vishal Seshadri B (CB.SC.U4AIE23260), Surya HA (CB.SC.U4AIE23267)** is a bonafide record of work carried out under my guidance at the **School of Artificial Intelligence**, Amrita Vishwa Vidyapeetham, Coimbatore, as part of the **IV Semester**, **Second Year** B.Tech curriculum in the course **"Intelligence of Biological Systems 2"** (**22BIO211**) and **"Design and Analysis of Algorithms"** (**22AIE212**).

Dr. Manoj Bhat K	**Dr. Vinith R**
Project Guide(s)	Project Guide(s)
Designation	Designation

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INTERNAL EXAMINER

AMRITA SCHOOL OF ARTIFICIAL INTELLIGENCE

AMRITA VISHWA VIDYAPEETHAM

COIMBATORE - 641 112

DECLARATION

We, **KS Venkatram (CB.SC.U4AIE23236), Surya HA (CB.SC.U4AIE23267), Sanggit Saaran KCS (CB.SC.U4AIE23247), Vishal Seshadri B (CB.SC.U4AIE23260)** hereby declare that this thesis entitled **"Prediction of Therapeutic Peptide using Deep Learning"**, is the record of the original work done by us under the guidance of **Dr. Manoj Bhat K**, Designation and **Dr. Vinith R**, Designation, Amrita School of Artificial Intelligence, Coimbatore. To the best of my knowledge this work has not formed the basis for the award of any degree/diploma/ associateship/fellowship/or a similar award to any candidate in any University.

Place: **Signature of the Student**

Date:

Dr. K.P. Soman
Professor and Dean
Amrita School of Artificial Intelligence
Amrita Vishwa Vidyapeetham

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AMRITA VISHWA VIDYAPEETHAM

K S Venkatram [CB.SC.U4AIE23236]

Sanggit Saaran KCS [CB.SC.U4AIE23247]

Vishal Seshadri B [CB.SC.U4AIE23260]

Surya HA [CB.SC.U4AIE23267]

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List of Abbreviations

DAA	Design and Analysis of Algorithms
CNN	Convolutional Neural Network
LSTM	Long Short-Term Memory
ProtBERT	Protein Bidirectional Encoder Representations from Transformers
AI	Artificial Intelligence
PTPD	Predicting Therapeutic Peptide by Deep Learning
TPpred-LE	Therapeutic Peptide prediction using Learned embeddings
PrMFTP	Prediction Model for Functional Therapeutic Peptides
APD3	Anti Microbial Peptide Database (Version 3)
iAMPpred	Intelligent Anti-Microbial Peptide Prediction
AntiBP2	Antibacterial Peptide Prediction Tool (Version 2)
MLACP	Machine Learning-based Anticancer Peptide predictor

Abstract

In the field of bioinformatics, the discovery of therapeutic peptides plays a vital role in the advancements of drug creation and enhancements. However traditional methods are often computationally expensive and lack scalability. Our project helps overcome these hindrances by introducing an innovative approach by combining a deep learning-based framework with DAA-based algorithmic optimization technique to address these challenges. We are utilizing a hybrid CNN-LSTM model which helps in identifying a user-entered sequence as therapeutic or not and then it further categorizes them into specific therapeutic categories which include Anti-Microbial, Anti-viral, Anti-fungal, anti-HIV, Anti Oxidant, anti-cancer, anti-protease inhibitors, wound healing peptides.

To enhance the computations of the models and the prediction's accuracy we have integrated DAA techniques, such as Rabin-Karp and Smith-Waterman for motif searching and sequence alignment. In Addition to this, we have incorporated advanced bioinformatics approaches which include ProtBERT embeddings, physicochemical feature extraction, and AlphaFold-based 3D structure prediction, providing comprehensive peptide categorization and prediction.

Molecular Docking simulations further find the therapeutic potential of the entered sequence by docking with the corresponding disease sequence, which then offers insights into peptide-target prediction.

Chapter 1

Introduction

1.1 Literature Survey

Recent developments in computational biology have significantly increased the discovery of functional properties of therapeutic peptides. Attique et al. (2020) conducted a detailed review of 41 studies, identifying the most effective machine learning classifiers and encoding schemes used in therapeutic peptide prediction. The authors of paper proposed a detailed classified framework that categorizes peptide prediction strategies based on algorithmic and biological characteristics. However, their study did not implement a novel predictive model and had inconsistency in performance due to a huge variance in their datasets.

Wu et al. (2019) developed PTPD, which is a deep learning-based model that used word2vec embeddings to predict anticancer peptides (ACPs) and virulent proteins. Their approach outperformed traditional methods, achieving impressive accuracies of 96%. Their model was highly focused on specific peptide classes and required more computational resources, doubting its scalability. Similarly, Lv et al. (2023) introduced TPpred-LE, which is a Transformer-based architecture capable of predicting 15 different therapeutic peptide functions. While the model achieved a good percentage of accuracy,

it demonstrated dependency on high quality labeled datasets and struggled to generalize when faced with unseen peptide types.

In an effort to classify multifunctional peptides, Yan et al. (2022) proposed PrMFTP, which leveraged multi-head self-attention mechanisms along with class-weight optimization. The model showed strong performance in identifying 22 multifunctional peptide types. However, its performance declined in the presence of highly imbalanced class distributions, and the training process was computationally intensive. Earlier, Xu et al. (2017) had presented a graph-based method called q-FP that modeled peptide sequences through graph indices, effectively classifying diverse range of peptide types. Although the approach was innovative it required additional validation and did not capture all sequence-level and structural peptide features.

Despite various developments in this field, existing literature reveals significant innovative gaps including limited standardization, lack of structural validation and were computationally expensive. To address these issues, our approach integrates ProtBERT-based transformer embeddings with CNN-BiLSTM architecture, enabling robust multi-class classification of therapeutic peptides. With this we extend our functionality beyond sequence-level prediction by integrating 3D structure modeling using OpenFold and peptide-disease docking simulations using AutoDock Vina.

1.2 Problem statement

Therapeutic Peptides play vital role in drug discovery and drug development. As new diseases are emerging day by day discovering and developing a potential therapeutic

peptide has become a challenge now. By following traditional methods, the steps for classifying therapeutic peptides from large biological datasets is computationally expensive and lacks precision due to its structural complexity and functional overlapping of recurring sequences. This delays the development of new medicinal drugs and treatments for diseases such as Cancer, Viral Infections, and Antibiotic-resistant Bacteria.

1.3 Objectives

- The primary focus is to develop a deep learning based approach for classifying therapeutic and non-therapeutic peptides based on biological features.
- Comparative representation of available peptide datasets, feature extraction methods, and classification models applied in therapeutic peptide research.
- Hierarchical arrangement of feature encoding methods depending on performance, underlining effective and successful schemes by means of DAA techniques.
- Development of an integrated framework for peptide–disease interaction analysis combining peptide classification, 3D structural generation (OpenFold), and molecular docking (AutoDock Vina).
- Deployment of a web-based platform with Postgres integration guarantees effective, scalable, and user-friendly access for real-time peptide screening and validation.

Chapter 2

Background

2.1 Therapeutic Peptides and their Biological Significance

Therapeutic peptides are short chains of amino acids which contains the functions of proteins which results in treating a disease with high specificity, minimal side effects, and diverse biological applications. These peptides are significantly better when compared with traditional pharmaceuticals, which helps in treating disease like Cancer, Infectious Diseases, Metabolic Disorders, and Autoimmune conditions. This ability help us to create peptide sequences which enables precise binding of biological targets.

2.2 Traditional Machine Learning Approaches

Traditional researches mostly involved traditional machine learning classifiers like Support Vector Machines (SVM), Random Forests (RF), and Decision Trees (DT) for peptide classification. These models relied on manually predefined features such as amino acid composition, physicochemical properties. Although they were effective upto some extent, they often failed to capture complex sequence patterns.

2.3 Deep Learning for Sequence Classification

Recent developments in deep learning models such as Convolutional Neural Networks (CNN), Long Short-Term Memory (LSTM), and Transformer-based architectures, have significantly improved peptide sequencing tasks and classification. These models learn sequences through their hierarchical representation, which showed major improvements in peptide related classification tasks when compared with traditional methods.

2.4 Therapeutic Peptide Databases and Tools

Biological Databases like Uni-Prot, Swissprot, APD3 provide peptide verified sequences and unverified sequences are provided by TrEMBL database. Tools such as iAMPpred, AntiBP2, and MLACP primarily focuses on single therapeutic functions. However, they often lack multi-class capability and structural validation, which affects the training of deep learning-based framework.

Chapter 3

Proposed Work

3.1 System Architecture

The proposed architecture is designed to classify and predict the peptide sequences through a multi-stage framework which consists of three different models. Model 1 performs binary classification to differentiate the user entered peptide sequence as therapeutic or non-therapeutic peptides. Sequences which are predicted as therapeutic are sent to Model 2, which classifies them into specific therapeutic classes such as Anti Bacterial, Anti Cancer, Anti Viral, etc. If a valid category is predicted, it returns the category name, then the sequence is forwarded to Model 3, which is termed the Disease Docker, which generates 3D structure of user entered sequence and simulates 3D structural interactions with a corresponding disease target by providing option to user on which target protein they would like to simulate for user specific needs. This workflow ensures the functional and structural validation of therapeutic peptides.

3.2 Sequence Pre-processing and Feature Engineering

3.2.1 Dataset Curation and Cleaning

Datasets are collected from verified biological databases like UniProt, Swiss-Prot and APD3 which provides peptide sequences. Repeated entries and ambiguous amino acids are removed to maintain consistency throughout the dataset. Sequences are then aligned through fixed-length padding and then converted into numerical representations using tokenization and other embedding techniques.

3.2.2 Feature Extraction Methods

The model uses a hybrid approach that includes both manual features and features that are gained through classical DAA Algorithms. Manually computed features include amino acid composition, dipeptide frequency, hydrophobicity, and instability index. Simultaneously, DAA based features are extracted using Smith Waterman Alignment, Rabin Karp Algorithm and Longest common subsequence. In addition to this predefined models such as ProtBERT are used, which preserve evolutionary and contextual information. The concatenated feature vectors provide rich biological and structural information to the classification models.

3.3 Deep Learning Models

3.3.1 Binary Classification: Therapeutic vs. Non-Therapeutic (Model 1)

Model 1 is implemented using a CNN-LSTM network that captures local motifs, sequential patterns through convolutional layers and sequential dependencies through LSTM. The model is trained on a balanced dataset using binary cross-entropy loss and evaluated with metrics such as accuracy and F1-score to ensure overall performance across all peptide types.

3.3.2 Multi-Class Therapeutic Classification (Model 2)

Model 2 does multi-class classification of therapeutic peptides. This model will predict the specific functional category of a given peptide sequence, such as Anti Bacterial, Anti Viral, Anti Cancer etc. It also uses ProtBERT embeddings to convert sequences into high-dimensional datasets that helps in capturing biological evolutionary patterns. These embeddings are then processed using a CNN-BiLSTM architecture, which effectively learns both local motifs and sequential patterns within the given peptide sequences. Then the model is trained on labeled data obtained from trusted biological databases such as UniProt, SwissProt etc and optimized with appropriate callback func-

tions to enhance the model's performance and prevent overfitting during early stages of model training. This deep learning-based framework provides a more scalable and more preciseful alternative to traditional tools that are limited to binary labeled predictions.

3.3.3 Disease Docker (Model 3)

Model 3 validates the predicted therapeutic peptide's functionality by simulating molecular interactions between therapeutic peptide and selected target proteins. First, the disease docker model generates 3D models of the user entered therapeutic peptide sequence using OpenFold which is Simpler version Google's Alphafold, then Based on the therapeutic category output from Model 2, a relevant disease target protein structure is retrieved from databases like PDB using an API call. Then the peptide and target protein are then processed using AutoDock Vina, which performs docking simulations and calculates binding affinity scores. A Higher negative binding scores indicate stronger interactions, which explicitly results in functional relevance of the predicted therapeutic peptide.

Chapter 4

Results and Discussion

Model 1 showed strong classification performance during training and testing validation. It began with a validation accuracy of 81.42% in the first epoch and showed a consistent increase during training. By the 10th epoch, the validation accuracy had reached around 87.56%, and the highest value was achieved around the 27th epoch with a validation accuracy of 89.49%. The model maintained high accuracy on upcoming epochs, indicating stability and generalization.

Model 2 was developed to classify therapeutic peptides into specific functional categories. The model achieved an overall accuracy of **81.16%**, with detailed class-wise performance metrics summarized in Table 4.1.

Category	**Precision**	**Recall**	**F1-score**	**Support**
Category1	0.78	0.65	0.71	5,000
Category2	0.75	0.80	0.78	5,000
Category3	0.66	0.75	0.70	5,000
Category4	0.65	1.00	0.79	5,000
Category5	0.98	0.96	0.97	5,000
Category6	0.96	0.46	0.62	5,000
Category7	0.95	0.92	0.94	5,000
Category8	0.95	0.96	0.96	5,000
			Accuracy	**0.81**
			Total Support	**40,000**

Table 4.1: Performance Metrics

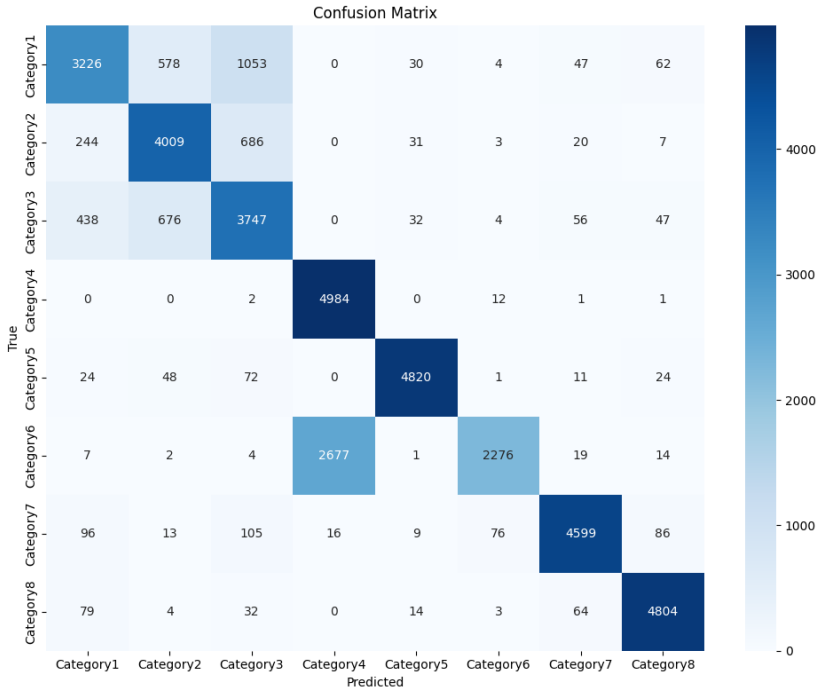

Figure 4.1: Confusion Matrix of Model 2

Figure 4.1 shows the confusion matrix for our therapeutic peptide classification model. Most of the predictions were accurate, especially for **Category4**, **Category5**, and **Category8**, where the model did a great job classifying and identifying the peptides. And the model showed some slight misclassification between **Category1**, **Category2**, and **Category3**, which might be due to similarities in their features. Overall, the model performs well, but there's still some scope to improve how it handles closely related categories.

Chapter 5

Conclusion

Our system consists of a deep learning framework named Therapeutic Peptide Finder to predict, classify, and simulate therapeutic peptides using a multi-model pipeline. This system combines advanced techniques from the field of bioinformatics and DAA Algorithms to achieve efficient and interpretable peptide-based predictions. The first stage focuses on distinguishing therapeutic from non-therapeutic peptides, by integrating ProtBERT embeddings and a hybrid CNN-BiLSTM architecture to capture sequence-level patterns. After successful classification, the pipeline proceeds to identify the peptide's therapeutic category such as Anti Bacterial, Anti Viral, Anti Cancer, or Anti Inflammatory. To enhance this multiclass model's performance, several DAA based optimization techniques were used, including recursive filtering and parallel computation ensuring that the model remains robust and scalable for large-scale datasets. The final model of the framework, the Disease Docker, simulates peptide and target protein interactions through 3D structure generation and molecular docking. This model allows users to select preferred target disease and visually explore the binding affinity between predicted peptides and target proteins using tools like OpenFold and AutoDock Vina. For the Tech Stack, backend is built using Django and PostgreSQL, ensures that previously processed sequences are stored and efficiently retrieved to re-

duce redundant computation. This framework enhances the speed and reliability of therapeutic peptide discovery. Overall, this work contributes significantly to the field by showing that combining deep learning with algorithmic can lead to more intelligent, automated, solutions in therapeutic peptide research.

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